*Name- Abhinav.A.Shejul
*Roll no.- 53
*Application no.- 2026SOC 4244

UNIT 1

1. Write a C program to swap two numbers using a temporary variable.

```
#include<stdio.h>
int main( )
{
   int a=5, b= 234567890, c;
   printf (" a= %d \n b= %d",a,b);
   c= a;
   a= b;
   b= c;
   printf ("\nafter swap");
   printf ("\n a = \n %d b= %d",a, b);
   return 0;
}
```

Output

```
a=5
 b=234567890
after swap
  a=234567890
  b=5
```

2. Write a program to calculate the area of a circle.

```
#include<stdio.h>
int main( )
{
 float radius, area;
 printf ("enter the radius = ");
 scanf ("%f",&radius );
 area= 3.14 *radius*radius;
 printf ("\n area = % f",area );
 return 0;
```

```
}
```

Output

```
Enter the radius= 4
Area=  50.24
```

3. write a programe to calculate the average of two numbers

```
#include<stdio.h>
int main( )
{
 float a, b, avg;
 printf ("give value of a= ");
 scanf ("%f",&a);
 printf ("\n give value of b= ");
 scanf ("%f",&b);
 avg= ( a+b)/2;
 printf (" average = %f", avg);
 return 0;
}
```

Output

```
give value of a= 5

 give value of b= 25
average = 15.000000
```

UNIT 2

1. Write a programme to check no. is even or odd .

```
#include<stdio.h>

int main( )

{

int a;

printf ("enter the no. ");
```

```
scanf("%d",&a);
if(a%2==0)
{
   printf ("is an even no. ");
}
else:
{
   printf ("is an odd no. ");

}
return 0;
}
```

Output

enter the no. 6899

is an odd no.

2. Write a programme to find multiplication of table using while loop.

```
#include<stdio.h>
int main()
{
int a,i=1;
printf("enter the no. to find table of ");
scanf("%d",&a);
while(i<=10)
{
```

```
printf("\n %d x %d = %d",a,i,a*i);
i++;
}
return 0;
}
```

Output

```
enter the no. to find table of 5

 5 x 1 = 5
 5 x 2 = 10
 5 x 3 = 15
 5 x 4 = 20
 5 x 5 = 25
 5 x 6 = 30
 5 x 7 = 35
 5 x 8 = 40
 5 x 9 = 45
 5 x 10 = 50
```

3. Write a programme to find no. is positive, negative or zero.

```
#include<stdio.h>
int main()
{
 int n;
   printf("Enter the no. ");
   scanf("%d",&n);
   if(n>0)
   {
      printf("\n no. is positive");
   }
   else if(n<0)
   {
      printf("\n no. is negative");
   }
   else
   {
      printf("\n no. is zero");
   }
   return 0;
```

}

Output
Enter the no. 5

no. is positive

UNIT 3

1. To write a C program to accept elements of integer, float, and character arrays from the user and display the value and corresponding memory address of each array element.

```
#include < stdio.h >
int main( )
{ int intArr [5];
 float floatArr [5];
 char charArr [5];
 int i;
 printf ("Enter 5 integer elements: ");
 for (i = 0; i < 5; i++)
 {
    scanf ("%d", & intArr [i]);
 }
 printf ("Enter 5 characters: ");
 for (i = 0; i < 5; i++)
 {
    scanf (" %c", &charArr [i]);
```

```
 }
 printf ("\nInteger  Array:\n");
 for (i = 0; i < 5; i++)
 {
    printf ("Value = %d\ tAddress  = %p\n", intArr  [i], (void *)&intArr  [i]);
 }
 printf ("\nFloat  Array:\n");
 for (i = 0; i < 5; i++)
 {
    printf ("Value = %.2f\ tAddress  = %p\n", floatArr [i], (void *)&floatArr  [i]);
 }
 printf ("\nCharacter  Array:\n");
 for (i = 0; i < 5; i++)
    { printf ("Value = %c\ tAddress  = %p\n", charArr  [i], (void *)&charArr  [i]);
    }
 return 0;
}
```

Output

```
Enter 5 integer elements: 5
9
6
2
7
```

Enter 5 characters: s

%

^

&

*

Integer Array:

Value = 5 Address = 0x7ffd1e574600

Value = 9 Address = 0x7ffd1e574604

Value = 6 Address = 0x7ffd1e574608

Value = 2 Address = 0x7ffd1e57460c

Value = 7 Address = 0x7ffd1e574610

Float Array:

Value = 0.00 Address = 0x7ffd1e5745e0

Value = 0.00 Address = 0x7ffd1e5745e4

Value = 0.00 Address = 0x7ffd1e5745e8

Value = 0.00 Address = 0x7ffd1e5745ec

Value = 0.00 Address = 0x7ffd1e5745f0

Character Array:

Value = s Address = 0x7ffd1e5745db

Value = % Address = 0x7ffd1e5745dc

Value = ^ Address = 0x7ffd1e5745dd

Value = & Address = 0x7ffd1e5745de

Value = * Address = 0x7ffd1e5745df

2. write a C program to accept the elements of a one-dimensional array and calculate the sum of all its elements.

```
#include < stdio.h >
int main( )
{
 int arr [100], n, i, sum = 0;
   printf ("Enter the number of elements: ");
   scanf ("%d", &n);
   printf ("Enter %d elements: ", n);
   for (i = 0; i < n; i++)
   { scanf ("%d", & arr [i]);
   }
   for (i = 0; i < n; i++)
      {
         sum = sum + arr [i];
      }
   printf ("Sum of the array elements = %d\n", sum);
   return 0;
}
```

Output

Enter the number of elements: 369

Enter 369 elements: thgrew

Sum of the array elements = 1611769023

3. write a programme read two matrices of the same order, perform matrix addition, and display the resultant matrix using a two-dimensional array.

```
#include < stdio.h >
int main( )
{
   int A[10][10],B[10][10],C[10][10];
   int rows, cols, i, j;
   printf ("Enter number of rows and columns: ");
   scanf ("%d %d", &rows, &cols);
   printf ("Enter elements of matrix A:\n");
   for (i = 0; i < rows; i++)
   {
      for (j = 0; j < cols; j++)
   {
      scanf ("%d", &A[ i][j]);
   }
   }
   printf ("Enter elements of matrix B:\n");
   for (i = 0; i < rows; i++)
      {
         for (j = 0; j < cols; j++)
            {
               scanf ("%d", &B[ i][j]);
```

```
            }
        }
    for (i = 0; i < rows; i++)
        { for (j = 0; j < cols; j++)
            { C[i][j] = A[i][j] + B[i][j];
            }
        }
    printf ("\nResultant  Matrix (A + B ):\n");
    for (i = 0; i < rows; i++)
        {
            for (j = 0; j < cols; j++)
            {
                printf ("%d\t", C[i][j]);
            }
            printf ("\n");
        }
    return 0;
}
```

Output

```
Enter number of rows and columns: 2 2
 Enter elements of matrix A: 1 2
 3 4
Enter elements of matrix B: 5 6
7 8
Resultant Matrix (A + B):
 6    8
 10  12
```